

from Book 13.1

Q1
i/ $2xy^2 - 3x^2 + 5y^2 - 6$

is a polynomial expression.

ii/ $3xy^{-2}$

Not a polynomial expression.

iii/ $\frac{2}{n+2}$

or

$$2 \times \frac{1}{n+2} \quad \text{or} \quad 2 \times n^{-1} + 2^{-1}$$

Hence it is not a polynomial.

Q3,

Reduce the following into lowest form.

i) $\frac{p^2 - 100}{p^2 + 10}$

$\Rightarrow$

~~$(a^2 - b^2) = (a+b)(a-b)$~~

~~$p^2 - 10^2$~~

~~$\frac{(p^2 - 10^2)}{p^2 + 10} = \frac{(p+10)(p-10)}{p^2 + 10}$~~

Here not division possible and

Not common so

write it on the mixed fraction

$\rightarrow$ Numerator number in middle

$\frac{p^2 - 100}{p^2 + 10}$

$\rightarrow$ and denominator

$p^2 + 10$

Now $p^2 + 10 \times p^2 - 100$. 1 times.

1 Quotient

$$\begin{array}{r} p^2 + 10 \overline{) p^2 - 100} \\ \underline{+ p^2 + 10} \\ -110 \end{array}$$

-110

$\rightarrow$ Remainder

$1 - \frac{110}{p^2 + 10}$

$p^2 + 10$

Quotient

Remainder in Numerator

$$ii) \frac{3ab - 3a^2}{3a^2 + 6ab + 3ab^2}$$

sol

$$\frac{\cancel{3a} (b-a)}{\cancel{3a}(a+2b+b^2)}$$

Ans

$$iii) \frac{(a-b)}{(a+b)} \times \frac{(a^2+ab)}{2a^2-2b^2}$$

sol

$$\frac{(a-b)}{(a+b)} \times \frac{a(a+b)}{2(a^2-b^2)}$$

$$\frac{\cancel{(a-b)}}{(a+b)} \times \frac{a \cancel{(a+b)}}{2 \cancel{(a-b)} \cancel{(a+b)}}$$

$$\frac{1}{a+b} \times \frac{a}{2} = \frac{a}{2(a+b)}$$

Ans